



*Cotswold
Archaeological
Trust Ltd*

Tel: Cirencester [redacted]

Fax: [redacted]

[redacted]
Dept. of Archaeology & Numismatics
National Museum of Wales
Cathays Park
Cardiff
South Glamorgan
CF1 3NP

4th May 1994

Dear [redacted]

GREAT HOUSE FARM, LLANDOUGH, SOUTH GLAMORGAN

Following our recent telephone conversation I write to inform you that Cotswold Archaeological Trust is conducting an excavation at the above site of a Roman and early medieval cemetery. We anticipate that something in the order of the 250 burials will be lifted.

Subject to the agreement of the legal landowner I would like to offer the human remains, artefact collection and site archive to the National Museum for long term curation.

I understand that you have no chosen box-size for human remains, although if you would like to suggest a preference (and supplier?) will do our best to meet it. There are one or two artefacts which will require conservation and X-raying. Is this a service you offer, and if so is there any charge?

Please feel free to visit the site again (we will be there to the beginning of June); indeed I would be keen to meet up with you and/or Mark Redknap to discuss the interpretation of the burials.

With kind regards.

Yours sincerely

DEPT OF ARCHAEOL & NUMIS

09 MAY 1994

& NUMIS

TRANSFER OF TITLE

You have kindly offered the following item(s) as a donation to the National Museum of Wales. In order to complete the transfer of ownership from you to the Museum, we would be grateful if you would sign the following statement.

We/I hereby declare that we/I are/am the legal owner of the following:

artefacts and bone considered important for research and future retention by the National Museum of Wales from the Cotswold Archaeological Trust excavations on the site known as Great House Farm, Llandough, South Glamorgan

We/I confirm that it is our/my wish to donate to the National Museum of Wales the above as an absolute and perpetual gift and that any copyright we/I hold in the gift is transferred to the National Museum of Wales. **Should any artefacts be placed on display in any museum Ideal Homes Wales Ltd. should be acknowledged as the donor.**

Signed:



Date: 28.7.94

Name(s) of Donor: IDEAL HOMES WALES LTD

Address: Llantrisant Business Park,
Llantrisant,
Mid Glam. CF7 8YP

Accession No: 95.56H

Other reference:



*Cotswold
Archaeological
Trust Ltd*

Tel: Cirencester [REDACTED]

Fax: [REDACTED] 1

[REDACTED]
Dept. of Archaeology & Numismatics
National Museum of Wales
Cathays Park
Cardiff
South Glamorgan
CF1 3NP

1st August 1994

Dear [REDACTED],

GREAT HOUSE FARM, LLANDOUGH, SOUTH GLAMORGAN

Thank you for your letter of 23rd July containing the photo of the cross in a different position. [REDACTED] has very kindly made her records available to us, and had mentioned about the photo.

Work on the site is dragging on somewhat and I fear the total count will approach 1000 (although only about 60% of the burials are complete and thus probably warrant detailed study).

I also enclose the document transferring title to the museum of all the artefacts from the site.

Kind regards.

Yours sincerely

[REDACTED]
[REDACTED]



**Cotswold
Archaeological
Trust**

Headquarters Building
Unit 9, Kemble Business Park
Cirencester
Glos. GL7 6BQ
Tel: [REDACTED]
Fax: [REDACTED]

[REDACTED]
National Museum of Wales
Cathays Park
Cardiff CF1 3NP

27 November 1997

Dear [REDACTED],

Llandough, Penarth

You should now be in receipt of four boxes of finds from the above site. The material includes animal bone, pottery, stone and metal finds. I have enclosed a list of all finds and contexts. Could you please acknowledge receipt of the material at your earliest convenience. Thank you.

Yours sincerely,

Emma Han

[REDACTED]
[REDACTED]

LLAN94 Llandough, Penarth Glass

Context Small Find

unstrat	203
unstrat, 3 bags	
001	64
1255	174
1422	157
B4	-
B101	43
B436	113
B477	67
B482	-
B638	205
B640	206

Slag

52	-
65	-
689	-
1015	96
B59	9
B69	11
B69	13
B87	23
B94, x2	27
B98	-
B101	39
B106	42
B108	-
B110	44
B110	47
B110	48
B110	51
B110	52
B110	53
B313	82
B313	84
B373	97
B373	98
B373	102
B373	103
B394	-
B428	111
B435	117
B442	119

Slag cont'd

B442	151
B478	147
B480	128
B498	-
B547	160
B552	162
B552	164
B552	165
B552	166
B552	176
B552	183
B605	-
B605	191
B762	223
B1022	-

Misc finds

B66	275	wood
B110	55	coffin wood
B251		coffin wood?
B251		leather
unstrat	118	worked bone
B69	14	bone pin

LLAN94 Llandough Pottery

Context	Small find
---------	------------

unstrat	
---------	--

?	209
---	-----

003	
-----	--

004	
-----	--

007	
-----	--

064	
-----	--

190	
-----	--

192	
-----	--

521	
-----	--

1235	
------	--

1255	
------	--

1257	125
------	-----

1422	
------	--

2255	252
------	-----

2379	
------	--

2387	261
------	-----

2590	
------	--

2611	
------	--

2632	
------	--

2642	
------	--

2647	
------	--

2652	
------	--

B2	
----	--

B3	
----	--

B5	
----	--

B8	
----	--

B10/11/19	
-----------	--

B15	
-----	--

B16	
-----	--

B18	
-----	--

B26/27	
--------	--

B29	
-----	--

B33	
-----	--

B34	
-----	--

B35	
-----	--

B39	
-----	--

B41/51	
--------	--

B42	
-----	--

B43	
-----	--

B46	
-----	--

B47	
-----	--

B50	
-----	--

B61	
-----	--

B66	
-----	--

Pottery cont'd

B78	
B80	
B84	
B87	
B89	
B96	
B105	
B106	
B108	
B110	
B111	
B116	
B117	
B120	
B121	
B122	
B132	
B134	
B135	
B138	
B157	
B159	
B178	
B180	
B190	
B194	
B195	
B210	
B221	
B228	
B243	
B254	
B260	
B296	
B320	
B324	
B272	
B306	
B333	
B337	
B341	
B345	
B373	99
B373	104
B373	105
B373	108
B414	
B422	

B425
B486
B512
B525
B534
B542
B545
B562
B582
B583
B592
B592
B601
B601
B605
B631
B638
B640
B679
B691
B753-6
B768
B796
B817
B879
B885
B932
B933
B979
B992
B1021

Pottery cont'd

188
190

192

208
213

Amphora sherds: 1255, B116, B517, B542, B864
Evaluation pottery: 160/007, 160/008, Test Pit 5

LLAN94 Llandough, Penarth Animal Bone

Context	Small find
---------	------------

unstrat	
---------	--

004	
-----	--

007	
-----	--

052	
-----	--

157	
-----	--

321	
-----	--

477	
-----	--

520	
-----	--

1016	
------	--

1422	179
------	-----

2255	
------	--

2632	
------	--

2648	
------	--

B2	
----	--

B4	
----	--

B5	
----	--

B18	
-----	--

B22	
-----	--

B35	
-----	--

B47	
-----	--

B54	
-----	--

B96	
-----	--

B122	
------	--

B131	
------	--

B134	
------	--

B138	
------	--

B178	
------	--

B190	
------	--

B319	
------	--

B337	
------	--

B363	101
------	-----

B422	
------	--

B446	
------	--

B542	
------	--

B546	
------	--

B562	
------	--

B605	
------	--

B716	217
------	-----

B768	
------	--

B796	247
------	-----

B833	235
------	-----

B847	
------	--

Stone finds

Context	small find	description
unstrat	123	flint
B175	69	flint
B442	114	flint
B442	145	flint
B483	133	flint
B593	186	flint
B1024	272	flint
unstrat		pebble
1016	95	pebble
1255		pebble
B18		pebble
B142	140	pebble
B190		pebble
B228		pebble
B306	85	pebble
B322		pebble
B383		pebble
B391	106	pebble
B442	150	pebble
B448, 480, 481		pebble
B456		pebble
B498		pebble
B512		pebble
B521		pebble
B533		pebble
B542		pebble
B547		pebble
B548	172	pebble
B550	163	pebble
B551	169	pebble
B551	170	pebble
B551	171	pebble
B560	175	pebble
B562		pebble
B570	181	pebble
B587		pebble
B589		pebble
B627	196	pebble
B627	197	pebble
B628		pebble
B643		pebble
B680	212	pebble
B691		pebble
B727	120-122	pebble

B728	218	pebble	Stone cont'd
B753-756		pebble	
B768		pebble	
B791	228	pebble	
B846	254	pebble	
B890	259	pebble	
B896	255	pebble	
B983	266	pebble	
B1021	271	pebble	
unstrat	8	clay marble	
1255	154	quartz	
2379		whetstone	
B31	5	whetstone	
B127	60	whetstone	
B511		stone	
B532		stone	
B796	246	chalk ball	
B802	231	chalk	
B843	239	chalk	
B843	240	chalk	
B843	288	chalk	
B855	248	chalk	

LLAN94 Llandough, Penarth metal finds

SF	Context	Description
195	B627	Fe nail
198	B627	Fe frag
202	unstrat	Cu alloy object
204	B635	Fe object
211	B666	Fe nail
214	B691	Fe nail
215	B711	Fe nail
224	B777	Fe nail
225	2067	Fe chain
229	2067	Fe nail
230	2067	Fe nail
241	B839	Fe nail
242	B839	Fe object
244	B796	Fe nail
251	2255	Fe nail + 3 studs
256	B868	Fe nail
260	2387	Fe nail
263	B945	Fe nail
265	B956	Fe object
269	2612	Fe frag
270	B1022	Fe nail
374?	B1027	Fe nail
	unstrat	Fe hook
	unstrat	Cu alloy thimble
	unstrat	lump
	unstrat	Cu alloy shoe buckle
	unstrat	Cu alloy object
	B4 (018)	Fe nail
	B35	Fe hobnails
	B60	Fe lump
	B87	Fe nails
	B133	Fe frag
	B198	button
	B227	Fe nail
	B233	Fe nail
	B303	Fe frag
	B319	Fe nail
	B321	Fe nail
	B364	Fe nail
	B482	Fe frag

LLAN 94 Llandough, Penarth metal finds

SF	Context	Description
3	121	Fe knife
10	B69	Fe nail
15	B69	Fe frag
16	B69	Fe frag
19	?	Cu alloy button
21	B75	Fe nail
24	B87	Fe nail
25	B87	Fe nail
30	B93	Fe frag
31	B94	Fe nail
33	B93	Fe nail
34	B93	Fe nail
36	B98	Fe frag
37	B101	Fe nail
38	B104	Fe nail
40	B104	Fe frag
41	B106	object
45	B110	Fe nail
46	B110	Fe nail
49	B110	Fe nail
54	B110	Fe nail
56	B110	Fe hobnail
57	B110	Fe object
59	B125	Fe hobnail
61	B110	slag
61	B110	Fe nail
65	B143	Fe nail
66	B143	Fe nail
68	B175	Fe nail
70	B195	Fe nail
73	B204	Fe frag
73	B204	Fe nail
74	B205	Fe nail
79	B273	Fe object
81	B294	Fe nail
86	B321	Fe nail
87	B313	Fe nail
90	B321	Fe nail
92	B326	Fe nail
107	B402	Fe nail
112	B428	Fe nail
121	B449	Fe nail
122	B449	Fe nail
127	B470	metal ball

130	B482	Fe frag
136	B489	Fe nail??
136	B470	metal ball??
137	B470	Fe nail
138	B449	Fe nail
139	B449	Fe nail
141	B449	Fe nail
142	B449	Fe nail
143	B449	Fe nail
144	B457	slag
148	1255	Fe nail
152	B514	metal ball
161	B552	Fe nail
167	B553	Fe lump
168	B547	Fe frag
173	B552	Fe frag
180	B552	Fe nail
184	B593	Fe nail
185	B593	Fe nail
194	B605	Fe nail

Palaeopathology Study Group
Rheumatology Unit
Department of Medicine
Bristol Royal Infirmary
BS2 8HW

National Museum of Wales
Cardiff

11.5.99

DEPT OF ARCHAEOLOGY

13 MAY 1999

& NUMISMATICS

Dear [REDACTED]

Further to our telephone conversation last Friday re. the curation of the skeletal material from Llandough, I feel that there are a few points that need clarifying.

We are delighted and very encouraged that action is being taken towards curating the human bone and we greatly appreciate your support on this matter.

I would like to draw your attention to the disarticulated human bone as this was not explicitly discussed on Friday and I am not sure if you are including it in your assessment for storage and further boxes. If you refer to point 6 in our proposal, the storage issues discussed here refer only to the articulated skeletons and do not take into account the twenty or so fruit boxes containing unwashed disarticulated bone. As I think we discussed, I will not be examining this material for my thesis, or the bone report and as far as washing is concerned it will take least priority. This is not to say that this material is of no value and we anticipate that it has excellent potential to form the basis of further postgraduate research at a later date. However it requires a different form of analysis to that which I am presently applying to the articulated specimens.

If the disarticulated material is to be kept, we will require a further fifty to seventy boxes in addition to the two hundred that is required for the articulated remains. Also, please note that these estimates have been worked out with the boxes specified in our proposal in mind and may be subject to change should you prefer to order boxes yourself.

You mentioned that you are intending to re-box skeletons that have already been washed and boxed. Is this a long term plan or is it something that you intend to do fairly soon? Also if it is a requirement of the museum that each skeleton should have proper card labels with them it would be useful to have some as soon as possible so that as I examine the skeletons I can exchange them for the paper labels that they currently have. The same applies to packaging should you feel that the large plastic bags tied with string that have been used so far are not sufficient. I feel that this is worth dealing with now as I will not have time to go back to boxes once skeletons have been examined.

Lastly, I am slightly concerned about archiving the collection to museum standards. As we discussed I am presently marking up each box with a site code, box number, skeleton numbers, and colour-coded stickers as appropriate. This information will be put onto an excel spreadsheet which will constitute the catalogue. These measures are designed purely for our purposes and I hope they will fit in with the system adopted at the museum in a satisfactory manner. This raises another concern of mine over extra curatorial work that may be incurred as a result of the skeletons becoming part of a museum collection. I am making every effort to ensure that the skeletons are archived and stored in an appropriate manner but as my PhD is my priority, I am only doing the bare minimum.

We are very happy with the way the project is progressing our end. We have acquired a warm room to be used as a temporary store for the skeletons and we plan to start supervised bone washing this summer.

I apologise if I am going over old ground in this letter and if these are matters that you have already taken on board. I am however very keen that things should continue to go smoothly. Once again many thanks for your help and support with this.

Yours sincerely,



cc. , University of Bristol
, University of Bristol
, Cotswold Archaeological Trust

AMGUEDDFEYDD AC ORIELAU CENEDLAETHOL CYMRU
NATIONAL MUSEUMS & GALLERIES OF WALES

MEMORANDUM

To : [REDACTED]

From: [REDACTED]

Date: 24th May 1999

[REDACTED]

Human Bone from Llandough

Finds from the Cotswold Archaeological Trust, including bone, were donated to NMGW by Ideal Homes in 1995. A decision on how much skeletal material to accept awaited advice from the bone specialist.

I now enclose -

- i). a copy of a report I requested on the human bone from Llandough, by [REDACTED] and [REDACTED], the Research Assistant who will be writing up the skeletal evidence.
- ii) A supporting letter from [REDACTED]
- iii) Collection Impact Assessment Form

Many early medieval cemeteries, such as that at Twyn-y-Capel recently excavated on Anglesey, have poor bone preservation. The Llandough collection is of international importance by virtue of its size, date and degree of preservation and rarity, and I support the considered view of the paper that **all** of the bone is retained by the Museum on the basis of the scientific grounds presented. The future multi-disciplinary research potential of such an assemblage is great, and while we have in the past considered selective retention of only the complete skeletons, I now believe that the disarticulated remains should also be retained. They fall outside the scope of the research by [REDACTED], and will form an important, unstudied element of the overall assemblage.

The assemblage and the data generated by on-going and future research programmes will have considerable impact on future display potential of our refurbished galleries, which could include one or more of the following themes: reconstructed skulls, data on the pre-Norman population of South Wales, comparisons with data on Anglo-Saxon cemetery assemblages, detailed grave plans, significance of freestanding crosses on such sites, the only early medieval hernia belt from pre-Norman Britain and the relationship between early monastic sites (such as Llandough) and princely sites (such as Dinas Powys).

I have also discussed the collection with [REDACTED] of BMNH. While the skeletal collection in BMNH from Cathedral Green, Winchester, is only a selection collection, (which she regrets), her recent involvement with the Hereford Cathedral ('Mappa Mundi') excavations has resulted in a decision to retain the whole assemblage, with the exception of the charnel remains which had already been reburied. The Hereford group is considered very important for being on the Welsh border, and will form an important comparable group to Llandough. BMNH also house the superb Poundbury material (the whole collection, and important as it is largely pre-Saxon), all the Cannington skeletons (about 400) and have just taken on all the

Abingdon skeletons (Saxon). [REDACTED] fully backs the proposal to house the complete Llandough assemblage as a reserve for the future, with great research potential which will not be satisfied by the current research programme.

I append a Collections Impact Assessment form for the Collection. I would be grateful if [REDACTED] could make arrangements to order boxes, so that [REDACTED] can box up those skeletons she is about to wash and study. [REDACTED] has also requested archival quality labels, so that she can undertake proper labelling for us.



Recommendation for the Curation of the Human Bone from Llandough, South Glamorgan.

Introduction

The future of the skeletal material from Llandough was addressed at a post-excavation meeting held on Friday 4th December 1998 at which representatives of the Cotswold Archaeological Trust, The National Museum of Wales, Cadw, and the University of Bristol were present. It was proposed by the National Museum of Wales that they would curate a selection of specimens recommended by the University of Bristol Palaeopathology Study Group as being the best candidates to keep and that the remainder of the assemblage would be reburied. Any storage containers that were required for these specimens would also be provided by the National Museum of Wales.

The human bone from Llandough constitutes a rare assemblage in a good enough state of preservation to yield large amounts of demographic and palaeopathological information. The potential of the entire assemblage for future research is far-reaching, and it is the opinion of the University of Bristol Palaeopathology Study Group that all of the material should be retained in preference to a selection for the reasons outlined below.

1. The rarity of the collection.

This is the only large skeletal collection to have been excavated in Wales and it also constitutes the largest Dark Age assemblage in Europe. It is considered to be of equal importance to other internationally recognised human bone assemblages such as Saint Peter's, Barton on Humber, the largest English collection of skeletons, ranging in date from 900-1850. As this is the first Welsh collection to be examined the data will be extremely valuable as a comparison with this and other sites from which skeletal material has been recovered.

2. The poor condition and incompleteness of some of the sample does not affect the potential of this material to provide much information.

2.1 In a preliminary report produced by the Cotswold Archaeological Trust the skeletons from the northern part of the site are described as being in the worst condition and the least complete. 434 skeletons from elsewhere on the site are said to be in 'a good state of preservation.'

A formal assessment of the material was carried out in 1996 by the Palaeopathology Study Group to determine its general condition and to estimate how this might influence the amount of information to be derived from the skeletons.

Sixteen articulated skeletons representing different categories of completeness and from different areas of the site were examined. Although some were found to be very fragmentary and in a fairly poor condition, this did not prevent the full range of assessments from being carried out and skeletons from all of the categories yielded enough information.

The further examination of the skeletons has recently been undertaken as a PhD thesis (3.2 below) and to date 30 individuals have been recorded. So far it has been possible to assign age and sex to all of the specimens, and in most cases sufficient measurements could be taken to enable stature estimation. Evidence of osteoarthritis, trauma and infection has also been observed.

2.2 Those skeletons that are in the worst condition and have the fewest bones present retain importance in their capacity to increase the overall sample size and therefore the amount of data available for statistical analysis. This allows population trends to be studied demographically and palaeopathologically both temporally and spatially. A large sample would be required to investigate the prevalence of disease by phase, sex, and age in order to ensure that there were sufficient individuals in each group to draw valid conclusions. Investigations of this sort may reveal trends through time that have the potential to be cross referenced to historical and archaeological records.

2.3 The more fragmentary and less complete skeletons are ideal for taking samples from for research. In the majority of cases only small bone fragments are required for specialist investigation such as stable isotope analysis (5.2 below). This is preferable to removing bone from the more complete and better preserved skeletons.

3. It is not possible to anticipate future research agendas.

3.1 With the exception of the occasional specimen that exhibits evidence of a rare pathology or, for example, the skeleton that was found with the belt/ truss, the only justification for retaining the skeletal material from Llandough is as a reference collection for future scientific research.

3.2 At present the Llandough material is being examined for a PhD thesis on the health and socio-economic status of the individuals. A core component of this research involves the examination of skeletons from other collections of a similar date to collect comparative data. Although recording sheets and databases can be consulted, it is not possible to retrieve all of the required information from these sources alone. The available data will not have been collected with the current research aims in mind, and a lack of standardisation for recording pathological changes can make it difficult or

impossible to reliably interpret or use data gathered by others. Without reference to the assemblages themselves, the potential usefulness of comparative research such as this can be extremely limited.

3.3 Given the difficulty of anticipating future research agendas, there are inherent problems in the determination of the appropriate selection criteria to apply to the collection. The selection of skeletons from an assemblage recovered in Winchester that are housed at the British Museum of Natural History were found to be inappropriate for a specialist study on arthritis since only bones from individuals that showed evidence for pathology had been retained (Rogers, pers. comm.). The osteologist is only rarely presented with the opportunity to carry out such research since the majority of skeletal assemblages are too small, and it is through this type of research that osteology can make its biggest contribution to archaeology. In any research requiring demographic information the individuals that are 'normal' are just as important as those selected for their 'unusual' traits.

3.4 The amount of information that one can expect individual skeletal remains to yield is subject to their condition and completeness at the time of their examination. The original skeletal remains are subject to a large number of selection processes, ranging from site taphonomy to the conditions of excavation, washing and post-processing, from the moment of their interment. These selection processes constantly reduce the potential amount of information which can be recovered from any individual, and part of the value of a collection such as this is therefore the range and number of specimens it contains. To select individual skeletons from the Llandough collection for curation by the National Museum of Wales will constitute a further reduction of the amount of information available for future research.

4 Future multi-disciplinary research potential of the sample from Llandough

4.1 Now that the examination of the 3,000 skeletons from Saint Peter's, Barton On Humber has been completed, there are 5 new grant applications being sent to the Wellcome Trust for further research, with other grant applications pending. It is expected that the Llandough collection will generate similar grant applications, and in the light of the volume of scientific research that has been generated since the examination of the Barton On Humber material was completed, it is estimated that an assemblage of the size of that from Llandough also has the potential to generate considerable amounts of future research after initial examination has been completed. Current research topics on the Barton On Humber material include:

- DNA analysis on all individuals aged between 17 and 25 years dating between 15th and 17th centuries for the occurrence of plague. This is being carried out by [REDACTED] from [REDACTED]

- Dietary research through stable isotope analysis carried out by [REDACTED] from the Department of Chemistry, University of Bristol.
- A clinical study into the natural history of fractures by members of the Medical school, University of Bristol.

4.2 Large skeletal reference collections are vital for enabling anthropologists to develop new analytical and investigative methodologies, demonstrated by a thesis evaluating a new ageing technique on the material from Saint Peter's Barton On Humber by [REDACTED].

4.3 Further research will add considerably to our understanding and knowledge of the poorly documented early medieval period in Wales.

4.4 The advancement of scientific techniques that can be applied to human bone may lead to new discoveries that we can't in our present state of knowledge anticipate.

5. Current proposals to use the sample from Llandough

Interest has already been expressed in using the Llandough material for future research. This has been expressed at both a local and international level.

5.1 [REDACTED], from the Smithsonian Institute, Washington is proposing to carry out research on health in Medieval Britain. For this he is selecting a number of Medieval sites across Britain, among which the Llandough collection is important as it is the only assemblage that represents Medieval Wales. A research association with the Smithsonian Institute may attract more interest in the assemblage on an international scale and would set the National Museum of Wales alongside other internationally recognised institutions where other important skeletal assemblages are kept.

5.2 Llandough is currently on a grant proposal for post-doctoral research into stable isotope analysis in the Department of Chemistry, University of Bristol.

6. Storage issues

The skeletons are currently in storage at the University of Bristol. About 400 are in boxes that are not suitable for long term storage. We recommend that they are stored in boxes measuring 460mm x 310mm x 150mm. These boxes are of sufficient size, stack well and are convenient to handle. They are available from a supplier in Bristol and the price is set out below;

Price per box = £2.55

Number of boxes required = 200

Total cost = £510.00



AMGUEDDFYEDD AC ORIELAU CENEDLAETHOL CYMRU
NATIONAL MUSEUMS & GALLERIES OF WALES

M E M O R A N D U M

To: [REDACTED]

From: [REDACTED]

Date: 19th July 1999

Copied to: [REDACTED], Medievalist

Re: Llandough Human Remains

The packaging costs for the human remains from Llandough have now been calculated.

I have created a new box size which meets the basic dimensions in terms of length and width recommended by [REDACTED] in her report. The boxes, are however deeper, to enable them to fit our standard roller racking. I have obtained a price on the basis that the order would be delivered to Bristol University rather than to Cardiff (the box manufacturer is based in Bristol).

External box dimensions: 460mm x 360mm x 240mm with a 100mm depth lid. (S.H. Fiske reference P36222).

Price £3.06 per box. We would require 200 boxes.

Total cost for boxes £612.00

We would also require a ream of acid free tissue, but could supply this from our existing supplies.

I would also require some shelving in the stores on which to place these boxes. The main stores are by far the best room in which to store these human remains, for the area is air-conditioned appropriately for bone. By storing the bone here we would also be able to use the cheaper cardboard boxes rather than having to commission air-tight boxes and use silica gel. The best place for more shelving would be on the wall where we currently have the Pontnewydd Cave artefact cabinet. Perhaps [REDACTED] could be asked to quote for building these shelves?

[REDACTED]

[REDACTED]

School of Conservation Sciences
University of Bournemouth
Weymouth House (WG22)
Talbot Campus
Fern Barrow Poole
Dorset BH12 5BB

Telephone: [REDACTED]
Email: [REDACTED]

[REDACTED]
Archaeology and Numismatics
National Museum and Gallery of Wales
Cathays Park
Cardiff, CF10 3NP

8.05.02

Dear [REDACTED]

I am writing to let you know that I have now moved to my new post as research assistant to [REDACTED] at the University of Bournemouth. I would also like to take this opportunity to update you on my progress with the Llandough project. Forgive me if some of this is repetitive but over the last few months I have spoken to several different people including yourself, [REDACTED] and [REDACTED].

As you know anthropological and palaeopathological examination of the articulated skeletons (representing a total of 801 individuals) was completed by Christmas last year. I have sent the basic demographic data to [REDACTED] along with a short report by myself for inclusion in the excavation report. I enclose a copy of my short report in case you have not already seen this.

Between Christmas and my re-location to Bournemouth at the end of last month, I have concentrated on finishing my PhD research on the collection and completing the curatorial tasks that have accompanied the project. I am now 3 quarters of the way through writing up my thesis and hope to submit later this summer. I hope to start writing the bone report late summer/early autumn.

[REDACTED] contacted me shortly after the New Year to say that you were planning to collect the material in the spring because financially this was a convenient time for the museum. I would like to stress that we have never voiced keenness to have the collection removed from Bristol immediately after completion of its primary documentation as [REDACTED] suggests in an email he sent [REDACTED] a couple of days ago. I expressed my concern that although all primary examination and documentation had been completed, I still required access to the collection to re-examine specimens, and complete the radiographic and photographic archives for my thesis. Ideally, the collection should also be available while I write up the bone report after my thesis has been submitted. This requirement is not however such a priority to me. Therefore, when I spoke to [REDACTED] about this we agreed that we would be in touch mid April to decide on a date.

Subsequent to this conversation my colleague and supervisor, [REDACTED] has also expressed her own feelings relating to the collection's re-location. [REDACTED] is concerned about the collection's removal prior to the completion of the bone report. [REDACTED] has played a pivotal role in the Llandough project ever since its arrival in Bristol in 1996 and quite naturally has her own feelings about it. Not only has she been my PhD supervisor but she has also provided the official support to the project on behalf of the University of Bristol. This has been increasingly so since the sad illness and death of Dr Rogers, originally my primary supervisor and who was originally approached about the project by [REDACTED]. The circumstances of Dr Roger's illness meant that she was increasingly unable to play an active role in the project and as such kept a low profile (although her continuing support and advice to me was considerable and continued up until her death at the end of last year). By default more than choice, [REDACTED] has played an increasing role in the project and this has gone beyond the call of duty. She

has provided storage space in her 'bone store' for the material and has invested a considerable amount of her own time in helping me process and manage the collection. Undertaking such a role has involved a lot of commitment on her behalf and it is understandable therefore, that we are surprised that her part in all of this has been somewhat overlooked despite her presence at meetings.

It is for this reason that a date has not been finalised for collection. I know that [REDACTED] is keen to talk to you first but I understand that you have been busy with an exhibition. I suggest that it would be useful to all of us if we arrange a meeting to discuss this.

All of the collection has now been moved into the bone store in the Department of Archaeology (during data collection part of it was housed in the Rheumatology Unit where I undertook the work), and as I have informed you, this has now passed into the full guardianship of [REDACTED]. Almost all of the collection has been re-boxed into the boxes that you provided for the project. I have set up a box catalogue system for the collection and this was completed by our bone washers who were paid under bone washing scheme that we operated last summer. All processing and cataloguing has been done with previous experience of managing a similar system for English Heritage for the Barton on Humber human skeletal collection. The catalogue also resides with [REDACTED] along with notes explaining the system and how the collection works.

I would be very grateful if you would let me know what your management strategy is for the collection once it has been moved back to Cardiff. I know we talked briefly about this on the 'phone. We have worked very hard to ensure that the collection is managed to a professional standard and is suitable for further research. So far, this has been very successful. You have provided considerable support to this and your decision to curate the collection and subsequent investment has meant that we have been able to realise the full potential of this unique research material. I am now keen that the final stage of our involvement in this project is completed with success and that all parties are satisfied.

Once it had been confirmed in 1998 that you would be retaining the collection, I wrote to you and spoke to [REDACTED] about the processing and curation of the material in Bristol to ask if you had any requirements or standards that needed to be met. I was concerned about the implications of a skeletal collection that came into a research environment with an uncertain future, and which, half way through processing became a collection to be retained by a museum. I am aware that you employ certain curatorial procedures for managing collections and have a number of guidelines that need to be met for this although you have not divulged any specifications for the Llandough collection despite my enquiries. [REDACTED] and myself have overseen the remainder of the processing and curation of the collection under the assumption that since this is the first large skeletal collection to be curated by the museum you have been happy to take our advice on its management. I am therefore very concerned that you have not shown much interest in coming to talk to us about this, despite several suggestions by myself that your colleagues spend an afternoon with us to familiarise themselves with the special requirements of a large osteological collection such as Llandough.

I am aware that you are very busy at the moment, but a large osteological collection has special curatorial requirements (which I doubt [REDACTED], with the best will in the world, is familiar with, not being an osteologist) if it is to remain useful and accessible for further research by professional osteologists. I would hate to think that all our efforts to set up the collection for future research will have been wasted and I would like to think that other osteologists who go to use the collection in the future will recognise it as a collection that has been successfully managed.

I feel it is necessary to make you aware of my feelings because I will be very disappointed if this project that we have all worked extremely hard to complete (despite some difficult circumstances), reaches an unsatisfactory conclusion. I hope you will not take any of this personally and will recognise as I do that because so many different parties are involved this has not been the easiest project. It is because of this that I want it to be remembered as a success and not a nightmare!

One last matter before I go. [REDACTED] approached me the other day about the 'hernia' belt from Llandough. He is very interested in it and is convinced that it s a chastity belt. This is because of a reference he came across that describes a brother at Much-Wenlock monastery (715-716 AD) with such an item. He gave me the following reference and I thought you would be interested. It is; [REDACTED] 1990 Religion and Literature in Western England 600-800 AD.

I expect I will be visiting the museum or will be contacting you again at some point because my new post involves researching peri- and post-mortem modification in prehistoric human bone.

I look forward to hearing from you



Short Report on the Llandough Human Skeletal Remains

Summary

A total of 801 discrete skeletons were received for examination and these were examined as part of a PhD thesis ([REDACTED], forthcoming) between 1998 and 2001. This work was carried out in the Rheumatology Unit, Division of Medicine, University of Bristol.

This short report discusses the condition and completeness of the skeletons and the population demography only. A full bone report is forthcoming.

Methods

Each specimen was laid out in anatomical position in the laboratory and an inventory of the elements that had survived was completed. Condition was determined as good, fair, or poor, depending on the amount of post-mortem damage that was present on all bone surfaces. Skeletons that generally had un-abraded joint ends and well preserved bone surfaces were recorded as good, while a poor skeleton presented considerable damage to most joint ends, as well as abraded bone surfaces. The amount of each skeleton present (completeness) was determined as a percentage out of 100, and this was assessed visually. Ages and sexes were estimated by employing standard anthropological techniques (Buikstra and Ubelaker, 1994; Cox and Mays, 2000).

Condition, Completeness and Fragmentation

Most skeletons were in a good or fair condition and 43% were between over 50% to 100% complete.

Number of Good Skeletons (%)	Number of Fair Skeletons (%)	Number of Poor Skeletons (%)
280 (35%)	320 (40%)	201 (25%)

Table 1. Condition of skeletons

	<15%	15-25%	>25-50%	>50-75%	>75%
No. (%)	58 (7%)	74 (9%)	323 (40%)	213 (27%)	133 (17%)

Table 2. Completeness of skeletons

Fragmentation was considerable across all condition and completeness categories. No skulls had survived complete and of the 6,372 major long bones that were present (humerus, radius, ulna, femur, tibia and fibula (rights and lefts)), 373 (6%) were intact, 1,077 (17%) were complete but broken, and 4,922 (77%) were incomplete, some represented by just a few fragments.

Age and Sex

Depending on which skeletal indicators had survived, each skeleton was assigned an age range. Adult skeletons are harder to age than child skeletons and therefore ages were assigned in 10 year bands up to 45 years. Age determination beyond 45 years is less accurate because the degenerative changes that occur at a regular rate up to this age become less distinct. For this population mature adults were categorised as 45+ or 60+ depending the preservation of appropriate elements. Although changes on the face the pubic symphysis and endocranial surface of the skull allow ages to be determined beyond 65 years, these elements did not survive in a good enough condition to be successfully applied in most cases in the mature adult age category. Narrower age bands could be assigned to children, especially for those that had dentitions surviving.

For this analysis the middle of each given adult age range was taken and rounded up where necessary. For example, an adult aged between 33 and 45 years, an age of 40 years was assigned. For children the middle of each age range was taken. These ages were then categorised in the following way;

Fetus

0	Perinate
0-5	young child (older than 0 and less than or equal to 5)
6-15	Child
16-25	Young adult
26-35	adult
36-45	middle adult
45+	mature adult (not including 45)
60+	mature adult (not including 60)

The accuracy with which age can be determined from a skeleton depends on the condition and completeness of the specimen. In some cases, the skeletons from Llandough presented a limited range of ageing traits and could not be assigned an age band but where possible were assigned to one of the above categories. To indicate that their assessment was less reliable, these individuals were given the following codes;

400 infant
500 child
600 young adult
900 adult
700 middle adult
800 mature adult

Individuals for which no age indicators had survived but whose bones had completed growth were categorised as adult (code; 100) and could belong to any of the above adult age categories.

Where possible sexes were determined as definite male, definite female, possible male or possible female. Sexing was not attempted on the children.

Sex	Age Range											Total (% of 801)
	Fetus (% of 801)	0 (% of 801)	0-5 (% of 801)	6-15 (% of 801)	16-25 (% of 801)	26-35 (% of 801)	36-45 (% of 801)	45+ (% of 801)	60+ (% of 801)	Adult (% of 801)		
Male (% of 801)					25 (3.1%)	48 (6.0%)	46 (5.7%)	43 (5.4%)	7 (0.9%)	11 (1.4%)	180 (22.5%)	
Female (% of 801)					31 (3.9%)	54 (6.7%)	33 (4.1%)	14 (1.7%)	11 (1.4%)	9 (1.1%)	152 (19.0%)	
? Male (% of 801)					6 (0.7%)	8 (1.0%)	10 (1.2%)	11 (1.4%)	2 (0.2%)	16 (2.0%)	53 (6.6%)	
?Female (% of 801)					7 (0.9%)	9 (1.1%)	5 (0.6%)	11 (1.4%)	1 (0.1%)	9 (1.1%)	42 (5.2%)	
Unknown (% of 801)	6 (0.7%)	13 (1.6%)	111 (13.9%)	84 (10.5%)	30 (3.7%)	8 (1.0%)	8 (1.0%)	7 (0.9%)	2 (0.2%)	105 (13.1%)	374 (46.7%)	
Total (% of 801)	6 (0.7%)	13 (1.6%)	111 (13.9%)	84 (10.5%)	99 (12.4%)	127 (15.9%)	102 (12.7%)	86 (10.7%)	23 (2.9%)	150 (18.7%)	801	

Table 3. Demographic profile

The demographic profile of the assemblage reflects a mixed group with a normal mortality curve for a population of this date and type (Waldron, 1995). Deaths peak in infancy, early adulthood, and later adulthood. The peak of deaths in later adulthood is perhaps slightly more gradual than usual. Deaths are more common among females in young adulthood (16-25) and adulthood (26-35) when parity is most likely. The number of individuals in the 60+ age category is low compared to those in the 45+ age category, and this may suggest that individuals tended not to live much beyond 45 years. It is possible however that this reflects a preservation bias rather than a true trend (ie. that few individuals had the appropriate traits available with which to determine an age beyond 45+). This requires further investigation.

A crude analysis shows that 26.7% of the total population is represented by children aged 15 years and younger and 73.3% is represented by adults aged over 15 years. This is similar to the assemblage from Cannington, Somerset (Rahtz et al. 2000). Unlike Cannington which has a higher prevalence of females than males, males exceed females at Llandough by about 10%. Although it was not possible to determine the sex of 160 (27.3% of all adults), adults, this is thought to be a fairly accurate trend since the number of individuals that could be sexed is higher (427 adult individuals, or 72.7% of all adults).

References

Buikstra, Jane E. and Ubelaker, Douglas H. (eds), 1994 *Standards For Data Collection From Human Skeletal Remains* Proceedings of a Seminar at The field Museum of Natural History Organised by Jonathan Haas. Arkansas Archaeological Survey, Fayetteville, Arkansas.

Cox, M. and Mays, S. 2000. *Human Osteology In Archaeology and Forensic Science* London Greenwich Medical Media

Rahtz, P. Hirst, S. and Wright, S.M. 2000. *Cannington Cemetery. Excavations 1962-63 of prehistoric, Roman, post-Roman, and later features at Cannington Park Quarry, near Bridgwater, Somerset.* Britannia Monograph Series No.17. Society for the Promotion of Roman Studies.

Waldron, T. 1994. *Counting The Dead. The Epidemiology of Skeletal Populations.* West Sussex, England John Wiley and Sons Ltd

Llandough Human Skeletal Remains Catalogue System

Some notes and guidelines to the collection.

08.05.02

by

[REDACTED]

1. Boxes

The Llandough skeleton boxes contain material that divides into three main groups;

1. Boxes containing **discrete articulated skeletons** examined for the bone report.
2. Boxes containing **bone groups** not examined for the bone report.
3. Boxes containing **disarticulated** and **unstratified** human bone, and **grave fills**, not examined for the bone report.

1.1 Boxes containing discrete articulated skeletons examined for the bone report.

There are two types of box that contain these;

1. **Cardboard long boxes**
2. Shorter and squarer cardboard boxes (referred to as the **new boxes**)

The former represents the original boxes that used to store the material. These contained the proportion of the collection that had been processed by CAT between 1994 and 1996 and were delivered to the University of Bristol in 1996. The latter were purchased by the National Museum of Wales following their decision to curate the collection end of 1998/beginning of 1999. These were provided for the un-processed material that was also delivered to the university in 1996 and was contained in grocery boxes.

There were enough new boxes to re-box some of the skeletons contained in long boxes. The long term plan of the National Museum of Wales is to have the entire collection stored in the new boxes.

The material divides into **183 new boxes** and **221 long boxes**. **The number of skeletons in each box** varies depending on preservation and whether there are adults or children present. Most boxes contain between 1 and 4 skeletons, although there are some that contain more. Each skeleton is contained in one or more bags bearing the site code and skeleton number. Each bag contains labels with the skeleton number and site code.

Most boxes described in this section contain the **discrete, articulated skeletons** that are presented in the summary bone report and will also be presented in the main bone report (in progress). This is with the exception of some **bone groups** that are boxed as discrete skeletons with a **SK number** (see below). This is because their status as bone groups was not confirmed until after analysis had been completed and numbers for the site report were finalised. Also, some had already been boxed with the discrete articulated specimens before full examination of the material by the present author began.

Although every attempt has been made to keep the number of bone groups that are boxed with articulated skeletons to a minimum, this was unavoidable in some cases. It is therefore important that anyone accessing the material checks that the specimen number they are looking for is a discrete skeleton and not a bone group by checking the box catalogue on the Llandough 1994 box catalogue data base (see points 2.1 and 2.2) and by checking the site report.

On the front and side of each box is written the **site code (Llan'94)**, the **skeleton number (SK)**, and the number assigned to that box, the **box number** (see point 2). Please note that each specimen number originally had the word 'burial' and not 'SK' (skeleton) proceeding it. At the beginning of the anthropological and palaeopathological analysis all specimen labels were changed so that 'Sk' proceeded each number. It is standard osteological practice to refer to specimens as skeletons and not burials since the latter denotes the whole burial context. This should not present a problem except to those using the CAT Access Graves database where the word burial and not Sk proceeds each specimen number.

On the front of most boxes are various stickers. These relate to the **sticker coding system** that was employed during primary examination of the material by the present author. Now that this has been completed, they no longer serve a purpose, but may be useful to be aware of. The sticker definitions are as follows;

White rectangle (usually on bottom right hand corner) = primary examination of all skeletons in the box is complete.

Yellow dot = the skeleton has teeth surviving

White donut = the skeleton is a child (under 18 years)

Red rectangle = the skeleton has been radio-carbon dated

Red dot = the skeleton has osteoarthritis

Blue dot = the skeleton has a fracture

Green dot = the skeleton has infection

Little notice should be taken of the last three. These relate to very few boxes. During the project, these stickers were phased out after it had been decided that they were difficult to apply to a collection for which the diagnosis of pathological conditions was on going.

Some of the new boxes have ***no stickers at all***. This is because they represent skeletons that were re-boxed from long boxes (which originally had stickers) at the end of the project when the sticker system was no longer required.

1.2 Boxes containing *bone groups* not examined for the bone report

Fourteen long boxes (numbered 1-14) contain bagged bone groups that were separated from the articulated remains during processing. These all have a Sk number and they are listed on the Llandough box catalogue database in the ***Bone Groups*** table (see point 2.2). Some of this material has been processed, although most of it has not.

1.3 Boxes containing *disarticulated* and *unstratified* human bone, and *grave fills*, not examined for the bone report

This material is contained in long boxes. These boxes have no number and are listed in separate tables in the box catalogue database (see below). All of it is unprocessed except for the grave fills which were checked for human bone.

1.4 Skeletons that Need Re-uniting

During preparation of the bone report, it was necessary to re-examine certain specimens by extracting the appropriate bones from their designated boxes. Some of these have not been returned and have been temporarily boxed into long boxes in numerical groups (eg 100's, 300's etc.). These are awaiting re-union. When re-uniting these elements It is important to note that ***in most cases it is the labels that are in the bags with them that denote the skeleton they belong to and not the skeleton number written on the outside of the bag***. These represent old bags that have been re-used.

2. The Llandough Catalogue

This catalogue records the location of the discrete skeletons, and the bone groups. It also lists the disarticulated, unstratified, and grave fills material.

The catalogue has been entered onto an Access database called **Llandough 1994 Box Catalogue**. This contains 4 tables called; **Bone Groups**, **Box Catalogue**, **Disarticulated and Unstratified**, and **Grave Fills etc.**

The bone groups and box catalogue tables list box numbers along side skeleton and bone group numbers. These were assigned to assist in locating this material. This system was necessary because the nature of the processing and the condition of the material meant that sequential numbers could not be assigned to the same box. It is therefore recommended that when specimens are looked out, this is done so by looking for the numbers of the boxes that contain them in the box catalogue. The disarticulated, unstratified and grave fills material tables list all material that fall into these groups but these have not been assigned box numbers.

All of these tables have been appended to this document. The box catalogue (2.1) has been printed out in skeleton number order and in box number order.

2.1 The Box Catalogue (the most important table)

This refers to the main collection that is described in point 1.1

The table contains 3 fields; skeleton number, box number (the number of the box that contains the skeleton), and context number. Not all skeleton numbers have context numbers entered with them. These can be looked up on the CAT Graves database if required.

The number 100 has been used to proceed all box numbers that are in the long boxes. It is important to know whether a box number denotes a long box or a new box because numbers will overlap. This is because the box catalogue was initially set up for the long boxes before plans were made to buy new ones. When the new boxes arrived, the box catalogue was started again from the number 1 because it was too complicated to continue with the long box numbers.

2.2 The Bone Groups Catalogue

This lists all bone groups that are boxed separately from the discrete skeletons and lists the number of the box that the material is in. This does not include bone groups that are stored in boxes with discrete skeletons (see point 1.1). This latter group is listed in the box catalogue table in the field called **Bone Group?**

2.3 The Grave Fills Catalogue

This mainly includes grave fills that were recovered from graves during excavation. Charcoal, coffin, possible cremated material and a few soil samples are also listed. Most of these have a context and/or associated

skeleton number. None of these have been processed, although they were all checked during bone washing at the University of Bristol for human bone that related to the skeleton associated with the grave.

2.4 The Disarticulated and Unstratified Catalogue

This lists all disarticulated and unstratified material. Some of this has a context number and/or skeleton number, but in many cases this information is absent. Miscellaneous material such as unlabelled bags containing bone and soil samples are also included.